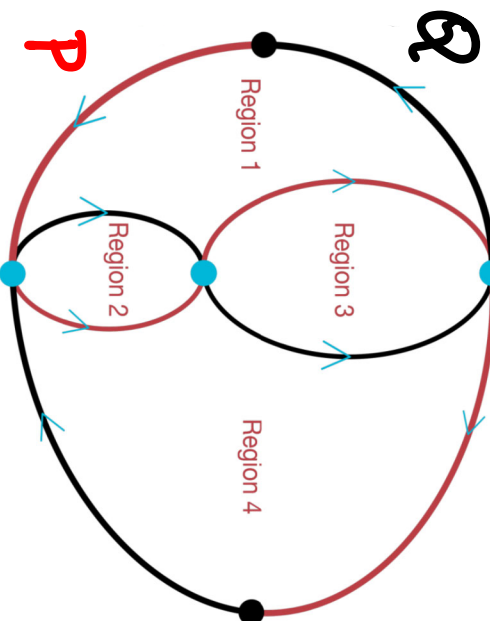


MINIMUM VOLUME HOMOTOPY IN DIMENSION ≥ 3

Bala Krishnamoorthy
Washington State University

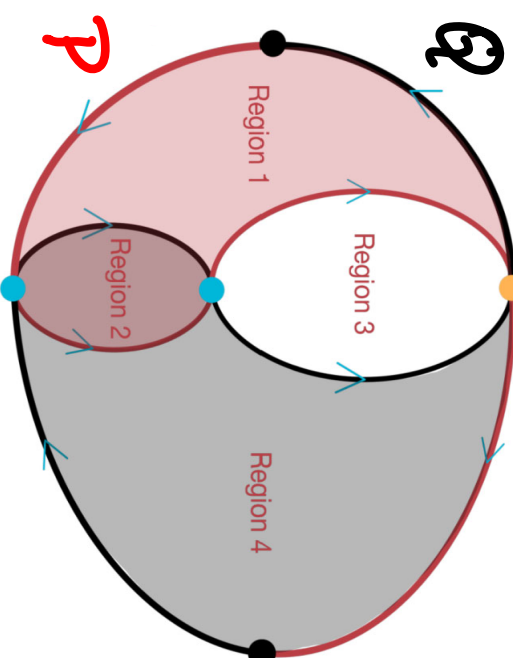
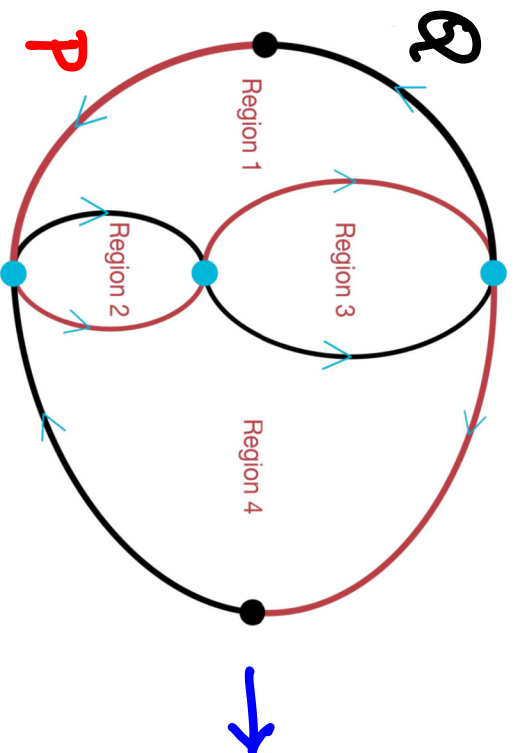
MIN. HOMOTOPY AREA OF CURVES (2D)

Chambers & Wang
(2013, 2019)



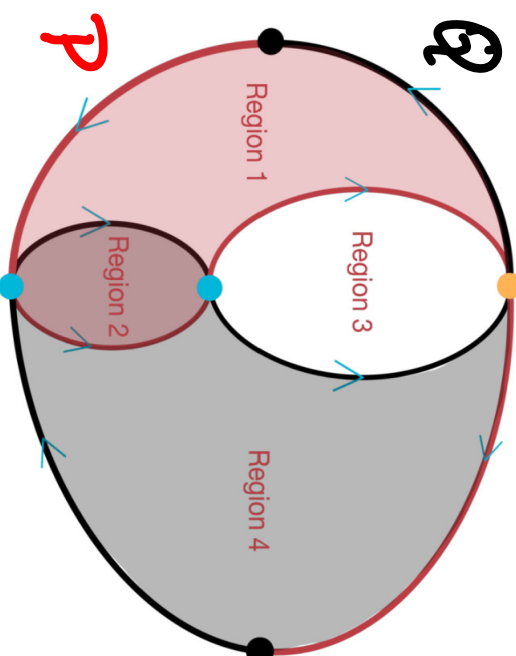
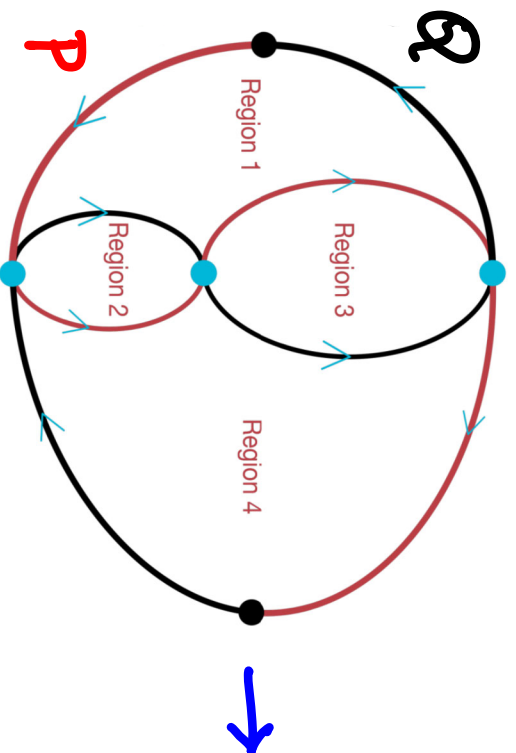
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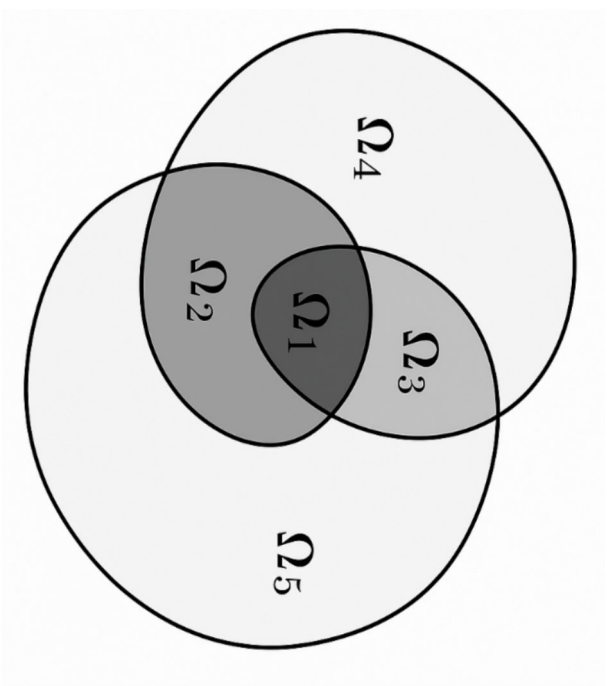


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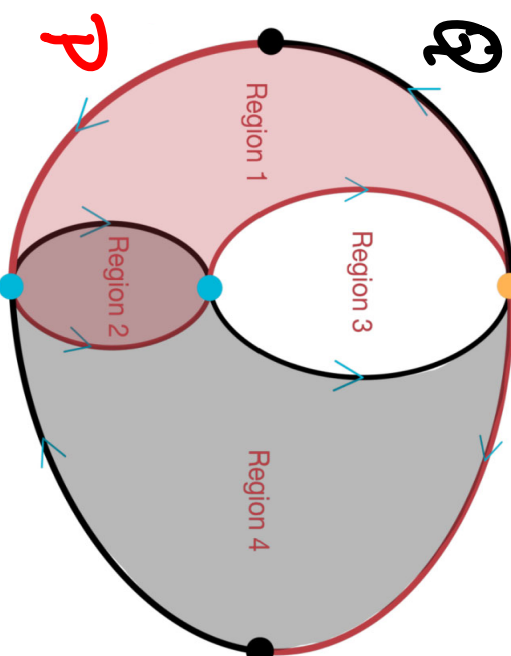
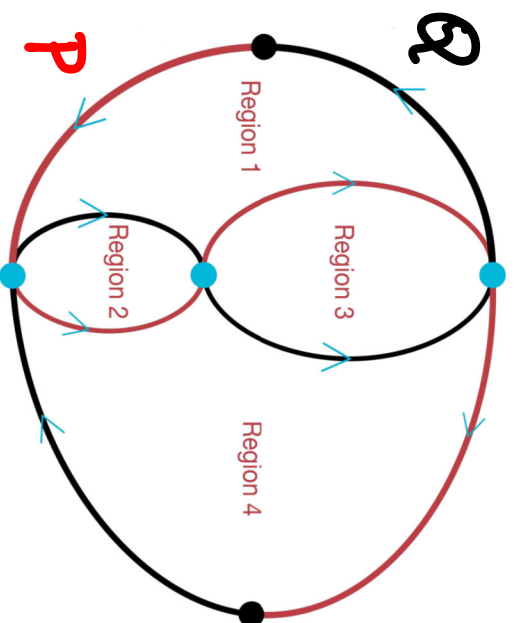


$\equiv$ closed curve, min area null homotopy

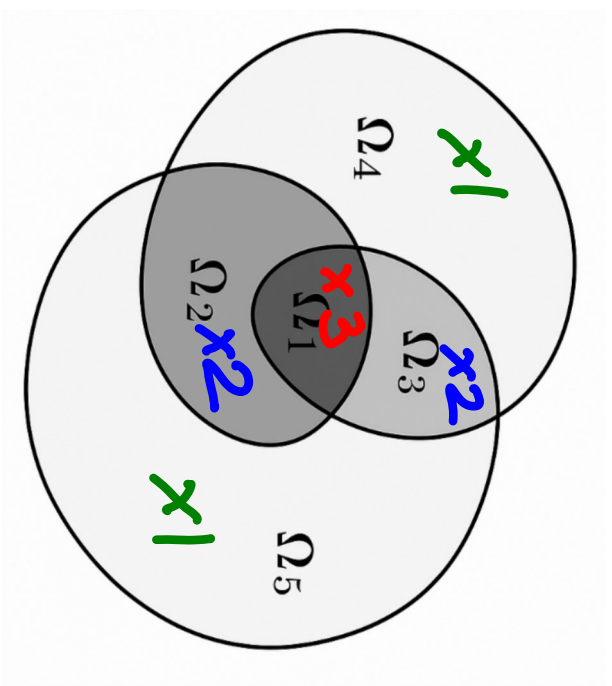


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- exact for sense-preserving homotopies

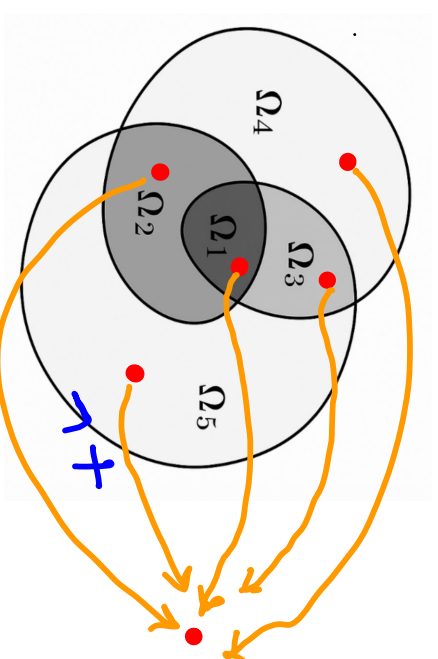
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- Blank word



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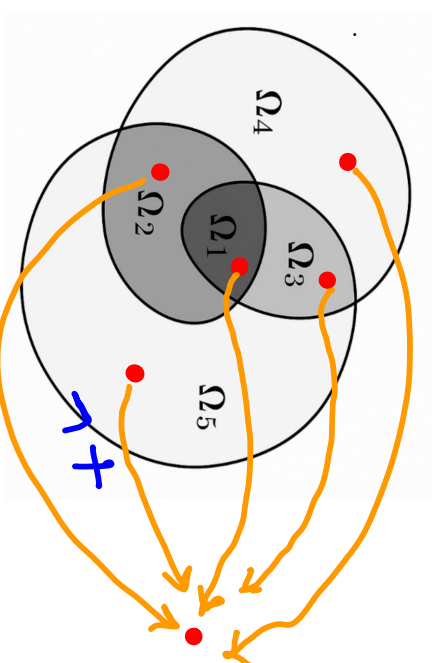
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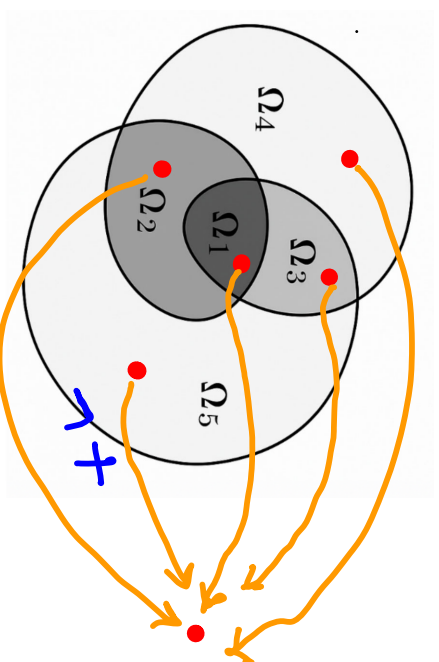
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* Nie (2014)

- algebraic word

* Chang, Fasy, McLoey, Millman, Wenk (2023)

- weighted cancellation, dynamic programming
- Blank $\equiv$ Nie words (under some conditions)



Min. Homotopy Volume in 3D?

? no "directed" walk along a surface?

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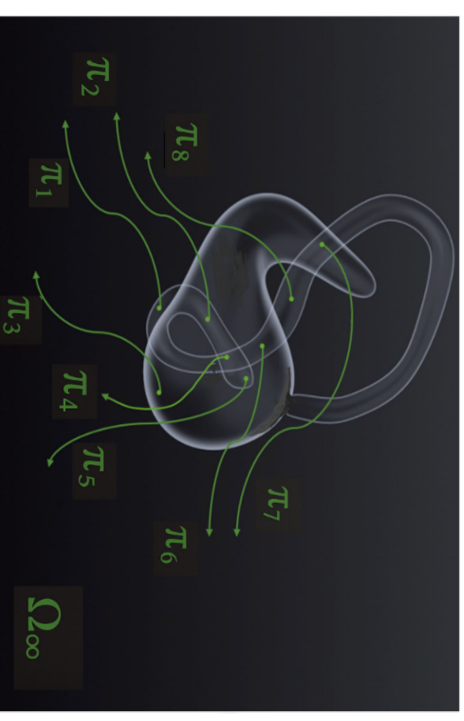
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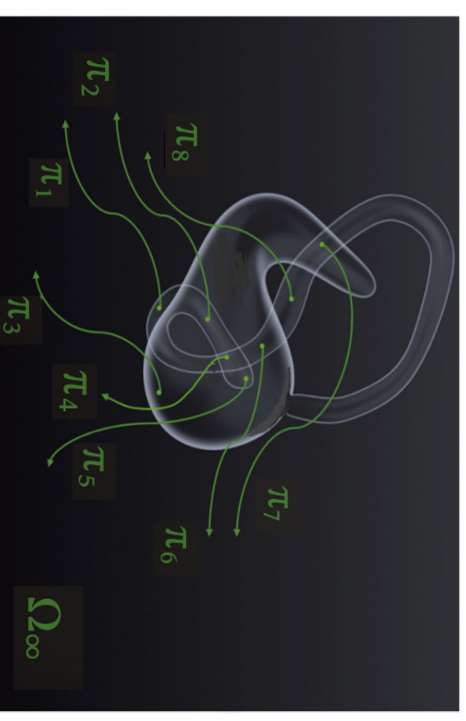


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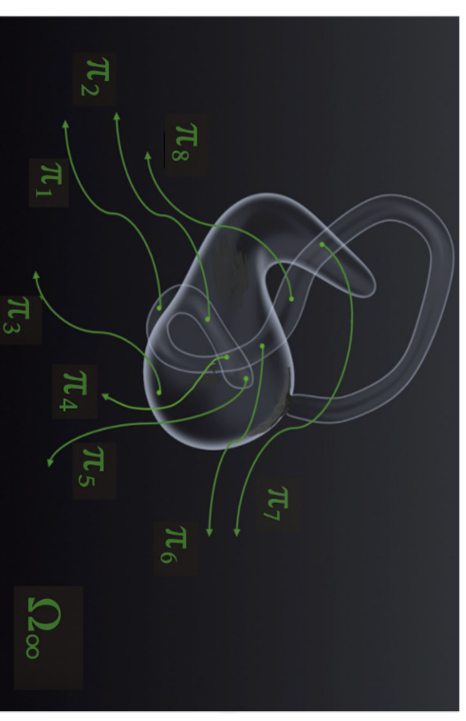
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? extension to d -sphere
for $d > 3$?



EXISTENCE OF MIN. HOM. AREA?

✓ Plateau's problem — existence of min. area surface spanning a Jordan curve
Douglas (1931), Rado (1930)

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✓ White (1983)
— existence under embedding assumptions